

Exercises

Category Theory, M1 MPRI, ENS Paris-Saclay

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1 Preorders and monoids as categories

Exercise 1 (Preorders as thin categories). A category C is *thin* if for any objects a, b of C , there is at most one morphism from a to b . A *preorder* is a set P equipped with a binary relation $\leq$ which is reflexive and transitive.

Prove that a small thin category is essentially the same thing as a preordered set. More precisely, define mutually inverse assignments between the collection of small thin categories and the collection of preorders.

Exercise 2 (Monoids as one-object categories). Prove that a small category with one object is essentially the same thing as a monoid.

2 Isos, epis, monos

Definition A. Let $f: a \rightarrow b$ be a morphism in a category C . For any object c of C , we define a function $f_c^*: C(b, c) \rightarrow C(a, c)$ by $f_c^*(h) := hf$ for any $h: b \rightarrow c$. We also define a function $f_c^+: C(c, a) \rightarrow C(c, b)$ by $f_c^+(g) := fg$ for any $g: c \rightarrow a$.

Exercise 3 (Isomorphisms in a category). Let $f: a \rightarrow b$ be a morphism in a category C . Prove that the following four properties are equivalent:

- (i) f is an isomorphism, i.e., there exists $g: b \rightarrow a$ such that $fg = \text{id}_b$ and $gf = \text{id}_a$;
- (ii) f has a left inverse and a right inverse, i.e., there exist $p: b \rightarrow a$ and $s: b \rightarrow a$ such that $fs = \text{id}_b$ and $pf = \text{id}_a$;
- (iii) for any object c of C , the function f_c^* is a bijection;
- (iv) for any object c of C , the function f_c^+ is a bijection.

Exercise 4 ($\text{Iso} \subsetneq \text{Mono} \cap \text{Epi}$). This exercise shows that the class of isomorphisms is included in the class of morphisms that are both mono and epi, and that this inclusion is strict in general.

- a. Prove that an isomorphism is both a monomorphism and an epimorphism.
- b. Prove that, in the category **Mon** of monoids and homomorphisms, the inclusion function $i: \mathbb{N} \hookrightarrow \mathbb{Z}$ is both a monomorphism and an epimorphism, but not an isomorphism.

Hint. Recall that a monoid homomorphism must preserve any existing inverses.

Exercise 5 (Split epimorphisms). Let C be a category. A morphism $p: a \rightarrow b$ in C is called a *split epimorphism* if it has a right inverse, i.e., if there exists $s: b \rightarrow a$ such that $ps = \text{id}_b$.

- Prove that a split epimorphism is an epimorphism.
- Give an example of a category C in which not every epimorphism is a split epimorphism.
- Prove that p is a split epimorphism if, and only if, for any object c of C , the function p_c^+ (see Definition A) is surjective.
- Show, without using the Axiom of Choice, that a function is surjective if, and only if, it is an epimorphism in the category **Set**.
- Recall that the Axiom of Choice (AC) states that, for any set $\mathcal{A}$ of non-empty sets, there exists a function f with domain $\mathcal{A}$ and such that, for every $A \in \mathcal{A}$, $f(A) \in A$. Prove that AC is equivalent to: in the category **Set**, every epimorphism is a split epimorphism.

3 Functors

Exercise 6. Let C be the category whose objects are the vector spaces $\mathbb{R}^d$ for each $d \in \mathbb{N}$, and whose morphisms are the linear transformations. Let $\mathbf{Mat}_{\mathbb{R}}$ be the category whose objects are natural numbers, whose morphisms from k to n are the $n \times k$ real matrices, and whose composition is given by matrix multiplication.

Define a functor F from $\mathbf{Mat}_{\mathbb{R}}$ to C which is an *isomorphism*, i.e., F is a bijection on objects and, for any objects n, k of $\mathbf{Mat}_{\mathbb{R}}$, the function $F: \mathbf{Mat}_{\mathbb{R}}(k, n) \rightarrow C(Fk, Fn)$ is a bijection.

Exercise 7. Let Σ be a set. Using Exercise 2, regard the monoid Σ^* as a one-object category. Denote by $\mathbf{Set}_f$ the category of finite sets and functions between them. Prove that the collection of functors $F: \Sigma^* \rightarrow \mathbf{Set}_f$ is in bijection with the collection of pairs (Q, δ) , where Q is a finite set and $\delta: \Sigma \times Q \rightarrow Q$ is a function.

Exercise 8. Let $F: C \rightarrow D$ be a functor.

- Prove that F preserves isomorphisms, i.e., if f is an iso in C , then Ff is an iso in D .
- Show that F does not necessarily preserve monomorphisms or epimorphisms.

Exercise 9. A *quiver* is a tuple $G = (V, E, d, c)$, where V and E are sets, and $d, c: E \rightrightarrows V$ are functions. For $a, b \in V$, a *path* from a to b in G , is a list $p = (a_n, f_n, a_{n-1}, f_{n-1}, \dots, a_1, f_1, a_0)$, for some $n \geq 0$, with $a_i \in V$ and $f_i \in E$ for each $0 \leq i \leq n$, such that $a_0 = a$, $a_n = b$, and, for each $0 < i \leq n$, $d(f_i) = a_{i-1}$ and $c(f_i) = a_i$; we write $p: a \rightarrow b$. The *category of paths* on G , PG , has set of objects V and set of morphisms the paths in G .

- Define an appropriate identity and composition on PG and show that PG is a category.
- Let C be a category. A *morphism* from G to C is a pair of functions $F_0: V \rightarrow \text{ob } C$ and $F_1: E \rightarrow \text{mor } C$ such that, for all $f \in E$, $F_1 f$ has domain $F_0(df)$ and codomain $F_0(cf)$. Prove that, for any morphism (F_0, F_1) from G to C , there is a unique functor $\bar{F}: PG \rightarrow C$ such that $\bar{F}a = F_0a$ for all $a \in V$ and $\bar{F}(ce, e, de) = F_1e$ for all $e \in E$.

Remark. The category PG is also called the *free category* on G .

4 Binary products and terminal objects

Exercise 10. Let P be a preorder, seen as a category (see Exercise 1).

- Let $t \in P$. Show that t is terminal if, and only if, t is maximal.
- Let $a, b, c \in P$. Show that c is a product of a and b if, and only if, c is a greatest lower bound (infimum) of a and b .

Exercise 11. a. Give an example of sets A and B such that the projection $A \times B \rightarrow B$ is not surjective.

- Conclude that product projections need not be epimorphisms.
- Show that the unique map to a terminal object need not be an epimorphism.

Definition B. Let I be an index set, C a category, a an object, and $(f_i: a \rightarrow b_i)_{i \in I}$ a family of morphisms with domain a . We say that the family $(f_i)_{i \in I}$ is *jointly monomorphic* if, for any pair of morphisms $g, h: x \rightrightarrows a$, if $f_i \circ g = f_i \circ h$ for every $i \in I$, then $g = h$.

We say that a jointly monomorphic family $(f_i: a \rightarrow b)_{i \in I}$ is *extremal* if, for any family of morphisms $(f'_i: a' \rightarrow b_i)_{i \in I}$ and any epimorphism $e: a \rightarrow a'$, if $f_i = f'_i \circ e$ for every $i \in I$, then e is an isomorphism.

Exercise 12. Let C be a category, let b_1, b_2 be objects in C and let $p_1: b_1 \times b_2 \rightarrow b_1$, $p_2: b_1 \times b_2 \rightarrow b_2$ be a product.

- Prove that the family (p_1, p_2) is jointly monomorphic and extremal.
- Let $(f_i: a \rightarrow b_i)_{i=1,2}$ be a family of morphisms with domain a . Prove that the family is jointly monomorphic if, and only if, $\langle f_1, f_2 \rangle$ is a monomorphism.

5 Binary coproducts and initial objects

Definition C. Let C be a category and a_1, a_2 objects of C . A pair of morphisms $i_1: a_1 \rightarrow b$ and $i_2: a_2 \rightarrow b$ is a *coproduct* if it is a product in C^{op} . An object a of C is *initial* if it is terminal in C^{op} .

Exercise 13. a. Write down a more concrete definition of ‘coproduct’ and of ‘initial’.

- Explain why coproducts and initial objects are unique up to isomorphism.
- Show that coproducts and an initial object exist in **Set**.
- What are a coproduct and initial object in a preorder?

Notation. A coproduct of two objects a_1, a_2 is denoted $a_1 + a_2$, and an initial object is denoted 0 . The copairing of two morphisms $f_1: a_1 \rightarrow c$ and $f_2: a_2 \rightarrow c$ is denoted $[f_1, f_2]: a_1 + a_2 \rightarrow c$. The functions $i_k: a_k \rightarrow a_1 + a_2$ are called *coprojections*, or also *injections* or *inclusions* (note that they are not necessarily monomorphisms, by the dual of Exercise 11).

Exercise 14. Let C be a category in which any two objects have a product and a coproduct.

- a. Show that, for any objects a, b, c , there exists a morphism $(a \times b) + (a \times c) \rightarrow a \times (b + c)$.
- b. Does there always exist an isomorphism?

Exercise 15. Consider the monoids $(\mathbb{N}, 0, +)$ and $(\{a, b\}^*, \epsilon, \cdot)$.

- a. Prove that $\{a, b\}^*$ is a coproduct of $\mathbb{N}$ and $\mathbb{N}$ in the category **Mon**.
- b. What is the coproduct of $\{a, b\}^*$ and $\mathbb{N}$ in **Mon**?
- c. What is the coproduct of $\mathbb{Z}$ and $\mathbb{Z}$ in the category of groups?

Exercise 16. Let Σ be a finite set. Consider the following category, which we denote $\Sigma/\mathbf{Mon}$:

- objects are pairs (M, f) , where M is a monoid, and $f: \Sigma \rightarrow M$ is a function from Σ to (the underlying set of) M ;
- a morphism $(M, f) \rightarrow (N, g)$ is a monoid homomorphism $h: M \rightarrow N$ such that $h \circ f = g$.

Show that $\Sigma/\mathbf{Mon}$ has an initial object.

Exercise 17. Let C be a category with a terminal object 1 and an initial object 0 . Prove that any morphism $f: 1 \rightarrow 0$ is an isomorphism.

6 Equalizers and coequalizers

Exercise 18. Prove that any equalizer is a monomorphism.

Exercise 19. Let $f, g: A \rightrightarrows B$ be two functions in **Set**. Define $\sim$ to be the smallest equivalence relation on B such that $f(a) \sim g(a)$ for every $a \in A$. Prove that the quotient map $e: B \rightarrow B/\sim$ is a coequalizer of f and g .

7 Pullbacks and pushouts

Definition D. The *walking span*, Λ , is the (poset) category with three objects and two non-identity morphisms as depicted on the left below. The *walking cospan*, V , is its opposite category, as depicted on the right below.



A *span* in a category C is a diagram of shape Λ , and a *cospan* is a diagram of shape V .

A *pushout*¹ is a colimit of a span, and a *pullback*² is a limit of a cospan.

Exercise 20. a. Write down explicit definitions, in terms of universal properties, of *pushout* and *pullback*.

¹In French: *somme amalgamée*.

²In French: *produit fibré*.

- b. Let $f_1: b_1 \rightarrow a$ and $f_2: b_2 \rightarrow a$ be morphisms in a category C . Suppose that a product $b_1 \times b_2$ of b_1 and b_2 exists in C , with projections $p_i: b_1 \times b_2 \rightarrow b_i$ for $i = 1, 2$, and that the pair of morphisms $f_1 p_1, f_2 p_2: b_1 \times b_2 \rightrightarrows a$ has an equalizer, $e: c \rightarrow b_1 \times b_2$. Prove that $(p_1 e, p_2 e)$ is a pullback of (f_1, f_2) .
- c. Let $f_1: B_1 \rightarrow A$ and $f_2: B_2 \rightarrow A$ be a span in **Set**. Show that there exists a pullback with summit $P := \{(b_1, b_2) \in B_1 \times B_2 \mid f_1(b_1) = f_2(b_2)\}$.
- d. Let $f_1: A \rightarrow B_1$ and $f_2: A \rightarrow B_2$ be a cospan in **Set**. Compute its pushout.

Exercise 21. Let C be a category containing the following morphisms.

$$\begin{array}{ccccc} a & \xrightarrow{p} & b & \xrightarrow{t} & e \\ \downarrow q & & \downarrow r & & \downarrow v \\ c & \xrightarrow{s} & d & \xrightarrow{u} & f \end{array}$$

Suppose that (r, t) is a pullback of (u, v) . Prove that (tp, q) is a pullback of (us, v) if, and only if, (p, q) is a pullback of (r, s) .

Exercise 22. Let C be a category in which all pullbacks exist.

- a. Prove that all equalizers exist in C .
- b. Suppose moreover that C has a terminal object. Prove that binary products exist in C .

Exercise 23. Prove that a morphism $f: a \rightarrow b$ is mono if, and only if, the square

$$\begin{array}{ccc} a & \xrightarrow{\text{id}_a} & a \\ \downarrow \text{id}_a & & \downarrow f \\ a & \xrightarrow{f} & b \end{array}$$

is a pullback.

8 Limits in general

Exercise 24. The aim of this exercise is to prove that *a category with small products and equalizers has small limits*.

Let C be a category and let $D: I \rightarrow C$ be a diagram in C , with I small. For each morphism t in I , write $d(t)$ for its domain and $c(t)$ for its codomain.

Suppose that the family of objects $(Di)_{i \in \text{ob } I}$ has a product $(p_i: P \rightarrow Di)_{i \in \text{ob } I}$ in C , and that the family of objects $(D(c(t)))_{t \in \text{mor } I}$ has a product $(q_t: Q \rightarrow D(c(t)))_{t \in \text{mor } I}$ in C .

Consider the morphisms $f, g: P \rightrightarrows Q$ defined as

$$f := \langle p_{c(t)} \rangle_{t \in \text{mor } I} \text{ and } g := \langle Dt \circ p_{d(t)} \rangle_{t \in \text{mor } I},$$

that is, $q_t \circ f = p_{c(t)}$ and $q_t \circ g = Dt \circ p_{d(t)}$ for every $t \in \text{mor } I$.

Suppose that $e: E \rightarrow P$ is an equalizer of the pair (f, g) .

Prove that $(p_i \circ e: E \rightarrow Di)_{i \in \text{ob } I}$ is a limit cone for D .

Definition E. A poset $(I, \leq)$ is *down-directed* if I is non-empty and, for any $a, b \in I$, there exists $c \in I$ such that $c \leq a$ and $c \leq b$. A diagram $D: I \rightarrow C$ is *projective* if I is a down-directed poset. A category C has *projective limits* if any projective diagram in C has a limit.

Exercise 25. The aim of this exercise is to prove that *a category with finite products and projective limits has all small limits*.

Let $(a_j)_{j \in J}$ be a family of objects in a category C . Define I to be the set of finite subsets of J , ordered by reverse inclusion, i.e., for $S, T \in I$, $S \leq T$ if, and only if $T \subseteq S$.

- a. Prove that $(I, \leq)$ is down-directed.

For each S in I , suppose that $(p_i^S: D_S \rightarrow a_i)_{i \in S}$ is a product of $(a_i)_{i \in S}$ in C .

- b. Show that whenever $S, T \in I$ and $S \leq T$, there exists a unique morphism $D_{ST}: D_S \rightarrow D_T$ in C such that, for any $i \in T$, we have $p_i^T \circ D_{ST} = p_i^S$.
- c. Conclude that $D: I \rightarrow C$ is a diagram.

Suppose that $(p_S: P \rightarrow D_S)_{S \in I}$ is a limit of D .

- d. Show that $(p_j^{\{j\}} \circ p_{\{j\}}: P \rightarrow a_j)_{j \in J}$ is a product of the family $(a_j)_{j \in J}$.

Exercise 26. Let C be a small category in which all small families have products. Prove that C is thin and complete.